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National University of Computer and Emerging Sciences, Lahore Campus

Course:	**Natural Language Processing**	**Course Code:**	**CS 535**
Program:	**MS(Data Science)**	**Semester:**	**Fall 2021**
Duration:	**40 Minutes**	**Total Marks:**	**18**
Paper Date:	**26-Nov-2021**	**Weight**	
Section:	**CS**	**Page(s):**	**4**
Exam:	**Quiz 2**		

Q1) Suppose we have embedding vectors of 3 dimensions for each word in the sentence "Energy minister tells petroleum dealers. Padding is applied to end and start of the sentence.

∅	**0.0**	**0.0**	**0.0**
energy	0.2	0.1	-0.3
minister	0.5	0.2	-0.3
tells	-0.1	-0.3	-0.2
petroleum	0.3	-0.3	0.1
dealers	0.2	-0.3	0.4
∅	**0.0**	**0.0**	**0.0**

(a) What will be output after applying following 2 filters to the above sentence. [2 Marks]

3	1	2
-1	2	1
1	1	-1

2	1	1
-1	-2	1
-1	1	-2

(b) Apply max pooling over time to the output obtained in part (a). [1 Mark]

Q2) What is the advantage of using CNN for text classification as compared to sequence based models like RNN and LSTM? [3 Marks]

>RNN introduced vanishing gradient problem while the LSTM fix that problem by additive forget gate property.
>RNN not able to capture long context while LSTM can capture by using selective info

Q3) In beam search for natural language generation, how do we decide when to stop the search? [3 Marks]

In beam search diff hypothesis generates end token on diff t stamp. So we'll continue until reach T stamp(where T is a predefine cut) or at least n completed hypothesis.

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Q4) What is the difference between beam search and greedy decoding. [3 Marks]

> In beam search on each step of decoder keep track K most probable partial translation called hypothesis
> size k 5-10
> hypothesis y1...yt is log of prob
> all -ve and higher score are better

> pick highest word as next word generation
> no way to undo

Q5)

We have encoder hidden states $h_1, \ldots, h_N \in \mathbb{R}^h$

On timestep t, we have decoder hidden state $s_t \in \mathbb{R}^h$

How will you calculate attention scores e^t. What will be dimensions of vector e^t

[2 Marks]

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Q6) Write prose and cons of Neural machine translation and statistical machine translation in the following table. [4 Marks]

Neural machine translation	**Neural machine translation**
Advantages: easy to debug, easy to control,	Advantages: better performance, better context use, no human effort involve, no individual components required, one network end to end
Disadvantages: complex sys, hundred of imp info involved, many separately design components, lost of feature engg involve, cost of human effort involved	Disadvantages: hard to debug difficult to control